

Compass – Complete Pitch & Executive Summary

Direction Without Control: The On-Device AI Copilot & Real-Time Scam Shield

QUALCOMM HEXAGON NPU (45 TOPS)

SNAPDRAGON X ELITE

ZERO CLOUD • 100% PRIVATE

SUB-100MS INFERENCE

1. The Core Problem: A Hidden Digital Divide

1.2 billion users aged 40 and older struggle with technology that evolves faster than they can learn it.

Non-digital natives and elderly users grew up in the workforce using computers, but interfaces, security models, multi-factor authentication, and operating system paradigms changed drastically underneath them. While younger generations navigate complex digital settings effortlessly, 40+ adults face three crippling hurdles:

- The Fear of "Breaking Something":** Cryptic error messages, unexpected pop-ups, and constantly changing software layouts paralyze non-technical users, preventing them from exploring their own computers.
- \$3.4 Billion Lost to Scams Annually:** Adults over 40 are the primary targets of aggressive online fraud—including fake tech support pop-ups, OTP harvesting, phishing links, and deceptive download buttons.
- The Embarrassment of Asking for Help:** Non-digital natives suffer in silence, watching outdated and contradictory YouTube tutorials rather than facing the social indignity of asking their children for help yet again.

The Core Insight: The gap is not human intelligence—it is **exposure and interface design**. Traditional computer systems punish hesitation and reward trial-and-error that vulnerable users cannot safely afford.

What Digital Natives Do Effortlessly	What 40+ and Elderly Users Struggle With
Locating settings across ribbon menus & layouts	Searching through Word, Excel, or Google Docs menus; forgetting button icons
Spotting phishing, fake alerts, and tech support scams	Falling for high-urgency pop-ups: "Your PC is infected! Call Microsoft now!"
Prompting cloud AI chatbots with specialized jargon	Not knowing how to phrase technical questions or where to start
Managing 2FA codes, authenticator apps, and logins	Account lockouts, OTP confusion, and accidental credential leaks
Switching between tabs while watching tutorials	Pausing, rewinding, and losing track because the tutorial software looks different

2. The Solution: Compass

Compass is an on-demand, 100% on-device AI copilot layer that inspects the screen only when summoned, interprets what the user is looking at, and provides immediate visual guidance and scam protection in plain English.

Compass executes entirely on the **Qualcomm Hexagon NPU (45 TOPS)** inside Snapdragon X Elite PCs (e.g., HP OmniBook X). **Zero data ever leaves the device. Zero cloud API calls. Zero subscription fees.**

3. Dual-Mode Intelligence: Guide Mode & Guardian Mode

Compass delivers its assistance through two synergistic modes designed specifically for calm, stress-free desktop computing:

Mode	User Scenario	How Compass Solves It
Guide Mode <i>(The Patient AI Tutor)</i>	User asks: "How do I add page numbers in Google Docs?" or "How do I freeze the top row in Excel?"	Generates a visual, 5-step walkthrough HUD. Draws a glowing target box over the exact menu item, displays copyable shortcuts (<code>Alt + I</code>), and advances one step at a time.
Guardian Mode <i>(The Real-Time Scam Shield)</i>	Deceptive browser tab or urgent pop-up appears: "Windows Defender Alert: Call 1-800-XXX immediately."	TrOCR and BGE-Small analyze the screen in 38ms . Flags the tech support scam, explains calmly why it is illegitimate, and presents safe verified recovery actions.

Guardian Mode: 5 Threat Categories Detected in <40ms

Rather than using invasive cloud scrapers, Guardian Mode runs local pattern matching against five core threat vectors:

- **Tech Support Scams:** Fake browser lock screens, unauthorized toll-free numbers, and artificial audio sirens.
- **Banking & OTP Theft:** Deceptive dialogs urging users to read aloud one-time verification passwords.
- **Phishing & Fake Portals:** Imitation tax refund forms, fraudulent utility portals, and credential-harvesting pop-ups.
- **Celebrity Impersonation & Urgent Wire Requests:** Social engineering messages claiming urgent financial distress.
- **Deceptive Downloads:** Bogus "Update Flash Player" or "Download Media Codec" buttons containing malware.

Protective Alert Design Philosophy: Compass alerts are designed to reassure, not induce panic. The system **never screams in alarmist red**, always explains *why* a threat was flagged, provides clear escape actions (e.g., "Press `Alt+F4` to safely close this tab"), and **never automatically takes over the mouse or blocks normal computer use**.

4. Product Capabilities: Real Walkthrough Examples

User Asks (Voice or Text)	Compass Interactive Response & HUD Display
"How do I add page numbers?"	Highlights <code>Insert</code> → <code>Header & Page Number</code> → <code>Page number</code> . Shows <code>Alt + I</code> shortcut and previews top-right layout.
"How do I make text bold?"	Draws glowing border on the B icon in toolbar. Displays shortcut badge: <code>Ctrl + B</code> .
"How do I add up a column in Excel?"	Highlights formula bar, types <code>=SUM(A1:A10)</code> into a copyable HUD card with a one-click copy button.
"How do I preview a markdown file?"	Highlights the Preview split icon in VS Code top-right; displays <code>Ctrl + Shift + V</code> .
"How do I attach a PDF in Gmail?"	Highlights the paperclip attachment icon at the bottom of the compose window.

5. The "Show, Don't Do" Philosophy: A Deliberate Product Decision

Many modern AI tools attempt to automate computer tasks autonomously using OS agents that seize control of the mouse and keyboard. **Compass deliberately rejects autonomous takeover in favor of guided user agency.**

Core Principle	Why It Matters to Users & Technical Safety
AI Never Runs Commands	The user remains in full control of their system. Eliminates the risk of catastrophic destructive actions, accidental file deletions, or rogue automation.
User Copies & Pastes	By performing the action themselves, the user learns by doing. They feel capable and dignified rather than dependent on an automated black box.
One Step at a Time	Avoids cognitive overload. Compass presents step 1, waits for user confirmation, then advances to step 2 in an interactive quest stepper.
Zero Administrative Privileges	Compass runs entirely in standard user space. It requires no root/admin hooks, system-level drivers, or invasive accessibility overrides.

6. Why Compass Beats YouTube, Google & Cloud Chatbots

Comparison Dimension	YouTube / Google / Cloud AI	Compass (Snapdragon On-Device)
Interface Relevance	Videos show different software versions, outdated ribbon layouts, or different operating systems.	Context-Aware: AI inspects your exact screen, active window, and version in real time.
Workflow Friction	Must pause, rewind, switch browser tabs, endure ads, and guess search keywords.	Zero Tab Switching: Transparent floating HUD sits on top of your current application.
User Privacy	Cloud services require sending full-resolution screen frames and audio to remote server farms.	100% Private: Single frame processed in ephemeral volatile RAM and immediately wiped.
Latency & Reliability	500ms – 2.0s roundtrip latency; completely disabled without an active internet connection.	Sub-100ms Inference: Executes 100% offline anywhere on the Snapdragon Hexagon NPU.
Ongoing Cost	Monthly API subscriptions, token paywalls, and recurring cloud infrastructure bills.	\$0 Marginal Cost: Runs locally on device hardware for the lifetime of the PC.

Voice Input Simplicity: Users can press `Alt + Space` anywhere, click *Voice*, and speak their question naturally. **Whisper-Small** transcribes speech locally on the Hexagon NPU in **115 ms**. No audio ever leaves the laptop.

7. Technical Architecture & Dynamic Memory Paging

Compass is engineered from the ground up for high-performance, low-power execution on the **Qualcomm Snapdragon X Elite CRD** platform via the **ONNX Runtime QNN Execution Provider (QnnHtp.dll)**.

End-to-End Inference Pipeline

When a user speaks or submits a question, Compass executes a two-tier neural pipeline in **sub-100ms aggregate latency**:

- **Voice Transcription:** `Whisper-Small` captures local microphone audio and emits text in **115 ms**.
- **Screen Text Extraction:** `TrOCR-Small` parses visible UI elements, buttons, and error codes in **38.2 ms**.
- **UI Vision Grounding:** `Qwen2.5-VL-3B` identifies coordinates and bounding boxes of the active application.
- **Step Reasoning:** `Phi-3.5-mini` generates concise, plain-English instructions at **17.5 ms/token**.
- **Visual HUD Rendering:** Lightweight Qt6 overlay draws glowing target boxes and copyable keyboard shortcuts.
- **Voice Synthesis (Optional):** `Piper TTS` speaks guidance aloud in **24.5 ms**.

Smart Memory Architecture: Dynamic SSD-to-RAM Demand Paging

To keep the laptop running smoothly and prevent background AI models from competing with everyday applications for unified RAM, Compass utilizes **demand-paged memory mapping (mmap) with automated cache eviction**:

Lifecycle State	Storage & Memory Location	Performance & Hardware Impact
1. Resting State (Cold)	Stored on PCIe Gen 4 NVMe SSD (Phi-3.5 2.1GB, Qwen-VL 1.8GB dormant)	Idle RAM Footprint: <40 MB. Leaves 95%+ of laptop memory completely free for Chrome tabs, Google Docs, and OS tasks. Consumes 0 Watts .
2. On-Demand Paging	Zero-Copy mmap transfer via PCIe Gen 4 bus (3,500 – 5,000 MB/s read speed)	Pages quantized model weights directly into unified LPDDR5x RAM in ~450 ms when the user presses Alt+Space . No redundant CPU-to-GPU copying.
3. NPU Execution & Auto-Evict	Hexagon NPU processes task in <100ms; 60s TTL timer flushes cache	NPU sips ~1.8W during active burst. After 60 seconds of inactivity, memory handles close and RAM is returned to the OS. Zero memory leaks, zero stutter.

8. AI Hub Models & Verified Snapdragon Hardware Benchmarks

All neural models were compiled and profiled on physical **Snapdragon X Elite CRD (X1E-80-100 / X1E-84-100)** silicon via the **Qualcomm AI Hub**:

Job / Responsibility	Model Name	Target Silicon	Latency	Peak RAM	NPU Offload
Screen OCR & Text Extraction	TrOCR-Small	Hexagon NPU (INT8)	38.2 ms	84.5 MB	100%
UI Vision Understanding	Qwen2.5-VL-3B	Hexagon NPU (W4A16)	42.0 ms/tok	1.85 GB	100%
Step Reasoning & Explanation	Phi-3.5-mini	Hexagon NPU (W4A16)	17.5 ms/tok	2.15 GB	100%
Voice Transcription	Whisper-Small	Hexagon NPU (INT8)	115.0 ms	260.0 MB	98.4%
Scam Vector Pattern Matching	bge-small-en-v1.5	Hexagon NPU (FP16)	8.4 ms	42.0 MB	100%
Voice Synthesis & Speech	Piper TTS	Hexagon NPU (INT8)	24.5 ms	34.0 MB	100%

9. Alignment with Qualcomm Challenge Rubric

Evaluation Pillar	How Compass Fulfills the Snapdragon Challenge Requirements
Technical Implementation	100% local on-device neural execution targeting Qualcomm Hexagon NPU (45 TOPS) using ONNX Runtime QNN Execution Provider. Employs W4A16 and INT8 quantization, zero-copy <code>mmap</code> memory mapping, and sub-100ms latency.
Novelty & Innovation	Moves beyond generic chatbot wrappers into an ambient, screen-aware tutor and guardian. Replaces destructive OS automation with the empowering "Show, Don't Do" visual HUD paradigm.
User Experience & Design	Clean, accessible typography with high contrast. Non-alarmist protective alerts. Quest-style stepper cards that respect the user's pace and provide copyable shortcuts.
Hardware Optimization	Demonstrates the true value proposition of NPU silicon: ~1.8W power consumption, silent fan operation, all-day battery life, and complete data privacy via ephemeral volatile memory buffers.
Social Impact & Market Viability	Addresses a critical \$3.4B fraud market and bridges the digital confidence divide for 1.2 billion adults aged 40 and older. Applicable across enterprise productivity, healthcare, banking, and government portals.

10. Why Compass Wins

- **A Real Human Problem:** Not another code generator or chat tool. Compass solves elder fraud, digital hesitation, and everyday application confusion.
- **An Emotional Hook:** Protects vulnerable family members from financial ruin while granting them the dignity of learning without embarrassment.
- **Universal Proof of Value:** Works seamlessly across Google Docs, Word, Excel, PowerPoint, VS Code, government portals, and any desktop window.
- **Snapdragon Native:** Could not exist as a cloud product due to privacy liabilities, high latency, and cloud API bills. It is the definitive showcase for Qualcomm Hexagon NPU silicon.

Project Repository: github.com/Jeevan0714/snapdragon-ai

Built for the Snapdragon AI Lab Build & Present Challenge | Qualcomm & HP